Sleep Medicine Reviews 15 (2011) 123–135

Contents lists available at ScienceDirect

Sleep Medicine Reviews

journal homepage: www.elsevier.com/locate/smrv

PHYSIOLOGICAL REVIEW

Adenosine, energy metabolism and sleep homeostasis

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ARTICLE INFO

Article history:
Received 23 April 2010
Received in revised form
30 June 2010
Accepted 30 June 2010
Available online 20 October 2010

Keywords:
Adenosine
Caffeine
Energy metabolism
Glia
NF-kB
Sleep
SD
Recovery sleep

SUMMARY

Adenosine is directly linked to the energy metabolism of cells. In the central nervous system (CNS) an increase in neuronal activity enhances energy consumption as well as extracellular adenosine concentrations. In most brain areas high extracellular adenosine concentrations, through A1 adenosine receptors, decrease neuronal activity and thus the need for energy. Adenosine may be a final common pathway for various sleep factors.

We have identified a relatively specific area, the basal forebrain (BF), which appears to be central in the regulation/execution of recovery sleep after sleep deprivation (SD), or prolonged wakefulness. Adenosine concentration increases in this area during SD, and this increase induces sleep while prevention of the increase during SD abolishes recovery sleep. The increase in adenosine is associated with local changes in energy metabolism as indicated by increases in levels of pyruvate and lactate and increased phosphorylation of AMP-activated protein kinase. The increases in adenosine and sleep are associated with intact cholinergic system since specific lesion of the BF cholinergic cells abolishes both. Whether adenosine during SD is produced by the cholinergic neurons or astrocytes associated with them remains to be explored.

An interesting, but so far unexplored question regards the relationship between the local, cortical regulation of sleep homeostasis and the global regulation of the state of sleep as executed by lower brain mechanisms, including the BF. The increase in adenosine concentration during SD also in cortical areas suggests that adenosine may have a role in the local regulation of sleep homeostasis. The core of sleep need is probably related to primitive functions of life, like energy metabolism. It can be noted that this assumption in no way excludes the possibility that later in evolution additional functions may have developed, e.g., related to complex neuronal network functions like memory and learning.

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Basic concepts of sleep

Introduction

Sleep, consisting of two main phases: non-rapid eye movement (NREM) and rapid eye movement (REM) sleep, is an essential part of the everyday life of animals as well as human beings. Despite intense research, the key function(s) of sleep remain unknown. Indirect evidence suggests that whatever the function of sleep is, or whatever was the reason that sleep was developed in the course of evolution, it is vitally important.

The rest–activity cycle, which often coincides with the diurnal changes in light–dark period, may be the ancestor or primitive state of the sleep–wake cycle. Since sleep is defined through electroencephalogram (EEG) recordings, by definition it exists only in animals with a relatively well-developed central nervous system (reptiles, birds, mammals). Recently the presence of "sleep", or sleep-like state, has been described also in simple organisms, e.g., cockroaches (*Leucophea maderae*), fruitflies (*Drosophila melanogaster*), zebra fish (*Danio rerio*) and, most recently, roundworms (*Caenorhabditis elegans*) (for review, see[1]). If we accept the notion that the rest–activity cycle represents primitive sleep in those species that do not have a well-developed CNS, we can say that all known animals sleep, which also signifies that "sleep" was developed early in the course of evolution. This suggests that the core of sleep need is related to primitive functions of life, e.g., energy metabolism.

Regulation of wakefulness

Cortical activation is maintained through the concerted activity of several cell groups, the waking system. The core of the waking

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1087-0792/$ – see front matter © 2010 Published by Elsevier Ltd.
doi:10.1016/j.smrv.2010.06.005

Abbreviations

AIR	A1 receptors
A2aR	A2a receptors
ACh	acetyl choline
ADA	adenosine deaminase
AK	adenosine kinase
AMPA	α-amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid
AMPK	AMP-activated protein kinase
ATP, ADP and AMP	adenosine tri-, di- and monophosphate
BDNF	brain-derived neurotrophic factor
BF	basal forebrain
cAMP	cyclic AMP
CNS	central nervous system
CNT	family concentrating transporters
CPT	Cyclopentyltheophylline
DNP	2, 4-dinitrophenol
EEG	electroencephalogram
ENT-family	equilibrative transporters
GABA	γ-amino-butyric acid
GHS	growth hormone secretagogue
GnRH	growth hormone-releasing factor
HDB	horizontal limb of diagonal band
IBMX	3-isobutyl-1-methylxanthine
ICV	intracerebroventricular
IL-1-β	interleukin-l-β
iNOS	inducible NO-synthase
KO	knock-out
LDT/PPT	laterodorsal and pedunculopontinetegmental nuclei
LPO	lateral preoptic area
LTP	long-term potentiation
MCPO	magnocellular preoptic area
MnPO	median preoptic area
NBTI	S-(4-nitrobenzyl)-6-thioinosine
NMDA	N-methyl-D-aspartic acid
NREM	non-rapid eye movement
PDE	phosphodiesterase
PDG2	prostaglandin D_2
PF-LHA	perifornical-lateral hypothalamic area
PKC	protein kinase C
PLC	phospholipase C
PnOc	oral pontine reticular formation
POAH	preoptic anterior hypothalamus
PRF	pontine reticular formation
REM	rapid eye movement
SAH	S-adenosylhomocysteine
saporin	IgG-saporin
SD	sleep deprivation
SI	substantia innominata
TNF	tumor necrosis factor
VA/VL	ventral anterior/ventral lateral
VLPO	ventrolateral preoptic area

system consists of the monoaminergic nuclei in the brainstem (raphe nuclei and locus coeruleus), the histaminergic tuberomammillary nuclei in the posterior hypothalamus, the hypocretin-containing cells in the lateral hypothalamus and the cholinergic nuclei in the brainstem (laterodorsal and pedunculopontine tegmental nuclei, LDT/PPT) and the BF.[2] In the current review the BF is defined as an area including three nuclei: substantia innominata (SI)/horizontal limb of diagonal band (HDB)/magnocellular preoptic area (MCPO). The cells of the waking system are active during waking and decrease their firing rate in NREM sleep.

As discussed in more detail in the following sections, we believe that the waking-active, presumably cholinergic, cells in the BF are the main targets for the vigilance state-modulating effects of adenosine. The BF cholinergic neurons send widespread projections to the cerebral cortex and to limbic structures without thalamic relay[3] and, to a lesser extent, to the hypothalamus.[2] The discharge rates of identified cholinergic neurons in the BF, as well as the cortical release of acetylcholine, are higher during wakefulness and REM sleep than during NREM sleep. Their discharge rate correlates positively with the power of fast gamma (30–60 Hz) and theta (4–8 Hz) activity on the EEG across sleep–wake states and correlates negatively with slow delta activity (1–4 Hz). Thus, cholinergic neurons in the BF are thought to stimulate high frequency gamma and theta activity during both waking and REM (for review, see[3]). Pharmacological experiments show that increased acetylcholine concentrations induce cortical EEG desynchronization, an effect that is antagonized by atropine, an acetylcholine receptor antagonist (for review, see[4]). Nonspecific lesions of the BF produce a decrease in cortical acetylcholine release, along with a decrease in cortical EEG activation.[5] Selective activation of the cholinergic BF neurons using neurotensin promotes EEG gamma activity and states of wakefulness and REM sleep.[6] Modulation of the activity of the cholinergic neurons thus provides a powerful tool for modification of vigilance states.

In the BF, cholinergic cells are intermingled with GABAergic and putative glutamatergic neurons which also project to the cortex.[3] A recent study indicates that a large population of identified GABAergic neurons (36%) discharged maximally during waking and REM sleep and in positive correlation with gamma EEG activity, while 28% discharged maximally during NREM sleep and in positive correlation with delta EEG activity.[7] Among the identified non-cholinergic/non-GABAergic (putative glutamatergic) cells 46% were maximally active during waking and REM sleep. Thus, it has been suggested that GABAergic and glutamatergic cells, whose discharge is positively correlated with gamma activity, along with cholinergic neurons, might stimulate cortical activation, albeit through different mechanisms and targeting different cortical neurons.[7]

It is important to understand that neural activity during waking is a central regulator of subsequent sleep. The concept of "use-dependent" sleep has been introduced and is supported by experimental evidence.[8,9] The quality of waking has profound effect on the quality of sleep: waking induced experimentally with N-methyl-D-aspartic acid (NMDA) receptor agonist induced high frequency theta activity and was followed by increased SWA activity in subsequent sleep, while similar duration of waking induced by α-amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid (AMPA)-receptor activation that did not induce high frequency theta during waking was not followed by increased SWA activity in sleep.[10]

Regulation of sleep

Sleep is promoted by the sleep-active cells in the hypothalamus. These neurons are most numerous in the preoptic anterior hypothalamus (POAH) (located medial to the cholinergic zone of the BF) where they comprise about 25% of the cells recorded *in vivo*. Lesions of the POAH in rats and cats produce significant reductions in sleep. Microinjections of somnogens such as adenosine and prostaglandin D_2 into the POAH significantly increase sleep, while intracerebroventricular (IVC) administration of interleukin-1-β (IL-1-β) and gh-releasing factor (GnRH) promote sleep via activation of preoptic neurons (for review, see[11]).

Segregated groups of sleep-active neurons are localized in the ventrolateral preoptic area (VLPO)[12] and median preoptic area (MnPO).[11] These cells express γ-amino-butyric acid (GABA), and in the VLPO neurons, GABA is co-localized with galanin. Data from c-fos studies as well as from electrophysiological and *in vivo* microdialysis studies suggest that VLPO neurons are active only during spontaneous sleep and recovery sleep, but not during periods of wakefulness preceding sleep, e.g., during SD,[11] suggesting that these neurons may not be involved in the mediation of the sleepiness associated with prolonged wakefulness, but rather in some aspect of maintenance or execution of sleep. However, the number of GABAergic neurons expressing c-Fos in the MnPO increase after 2 h SD as compared with spontaneous sleep and recovery sleep.[13] This is consistent with the data showing that c-fos immunoreactivity in MnPO increases with prolongation of SD[14] and that the gradual increase in the firing rates of most MnPO sleep-active neurons occurs prior to sleep onset.[11] These data suggest that MnPO is involved in the homeostatic aspects of sleep control.

Sleep propensity and homeostatic regulation of sleep

According to both experimental evidence and common knowledge, after prolonged wakefulness the degree of arousal, usually measured as EEG activation (EEG arousal), is decreased. Both the propensity to sleep and the intensity of delta EEG waves upon falling asleep have been demonstrated to be proportional to the duration of prior wakefulness.[15] The "extra sleep" that appears after prolonged wakefulness is called recovery sleep. The evidence from the electrophysiological experiments, performed in several species, and mathematically formalized in the two-process model of sleep regulation, is compelling, but the cellular mechanisms that could explain this phenomenon are still largely unknown.

Molecular correlates for sleep homeostasis: sleep factors

One possibility to explain how the propensity to sleep is accumulated during prolonged wakefulness is the introduction of the sleep factor theory. According to this theory a substance, or substances, sleep factors, accumulate in the brain during wakefulness. When the concentration of the factor(s) reaches a threshold value, sleep is induced. A sleep factor must meet several criteria: the concentration of the sleep factor must be higher during waking than during sleep, the concentration must decrease during sleep, and it must increase steadily during prolonged wakefulness. The substance must be able to inhibit neuronal activity: sleep is induced when activity in the waking-promoting neural networks decreases and activity in the sleep-promoting system increases. Many molecules meet some of these criteria, but very few meet most them. Adenosine, interleukin-1, TNF-alpha, prostaglandin D2 (PDG2) and gh-releasing factor (GnRH) belong to this category.

What is adenosine?

Introduction

Adenosine is a ubiquitous nucleoside which serves as a building block for nucleic acids and energy storage molecules, enzyme's substrate and neuromodulator of cellular activity.[16] In the CNS, adenosine is generally an inhibitory neuromodulator, inhibiting neuronal activity of both excitatory (e.g., cholinergic and glutamatergic) and inhibitory (e.g., GABAergic) neurons. However, adenosine is a more potent inhibitor of the release of excitatory neurotransmitters than of the inhibitory ones.

The rate of adenosine formation is regulated by neuronal activity.[17] The adenosine levels start to rise when the supply/demand ratio for oxygen decreases under circumstances like ischaemia, hypoxia and hypoperfusion. It has been suggested that this is a self-controlling neuroprotective mechanism to minimize cell damage: adenosine decreases cellular activity and thus decreases energy need.[18] Adenosine is also directly released upon neuronal activation (for review, see[19]), and it is co-released with several neurotransmitters (for review, see[20]).

Adenosine metabolism and transport

Through adenosine tri-, di- and monophosphate (ATP, ADP and AMP) adenosine is directly linked to the energy metabolism of cells (see Fig. 1). Intracellularly, the main pathway for adenosine formation is from AMP, catalyzed by the enzyme 5′-nucleotidase, extracellularly by the enzyme ecto-5′-nucleotidase. Intracellularly, adenosine can also be produced by hydrolysis of S-adenosylhomocysteine (SAH). In the periphery, the enzyme SAH-hydrolase forms SAH from S-adenosylmethionine. SAH-hydrolase is localized also in brain tissues, but the role of this pathway in the brain metabolism of adenosine is not clear. In the heart under good oxygenation a substantial part of adenosine is formed through SAH-hydrolase, but under hypoxia and ischaemia this pathway is not triggered (for review, see[21]). The inactivation of adenosine either back to AMP by the enzyme adenosine kinase (AK), or further to inosine by the enzyme adenosine deaminase (ADA) takes place intracellularly. The AK pathway is preferred at low adenosine concentrations, and ADA when adenosine concentrations are high.[21] The enzymes that metabolize adenosine undergo circadian variation with maximal level of enzymatic activity during the night and minimal during the day time.[22]

Significant diurnal variations in enzyme activities have been noted in several brain areas. In the BF, AK and both nucleotidases showed their lowest activity in the middle of the rest phase, 6 h after lights on, suggesting a low level of adenosine metabolism, both production and degradation, at this time point. In the cortex AK had a diurnal activity pattern similar to the BF, and the ecto-5′-

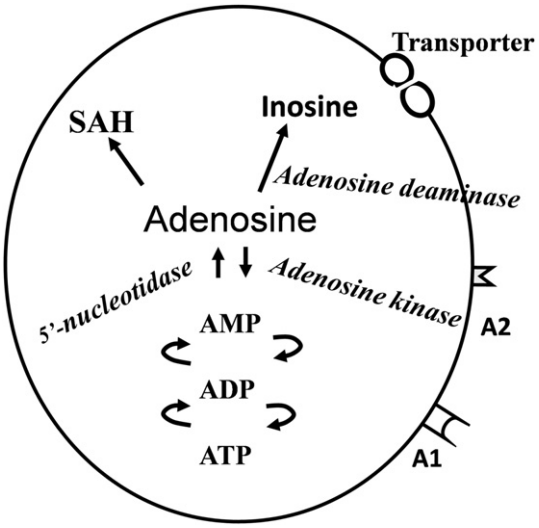


Fig. 1. Schematic of metabolic pathways of adenosine. Adenosine is formed from AMP intracellularly by the enzymes ATP-ase, ADP-ase and 5′-nucleotidase and extracellularly by the respective ectoenzymes. Adenosine kinase converts adenosine to AMP, while adenosine deaminase converts it to inosine. The third enzyme to metabolize adenosine is S-adenosylhomocysteine hydrolase, which converts adenosine to S-adenosylhomocysteine (SAH). Adenosine concentration between the intra and extracellular spaces is equilibrated by nucleoside transporters. Adenosine's effects on cells are mediated through specific receptors. Abbreviations: A1 = A1 adenosine receptor subtype; A2 = A2 adenosine receptor subtype; ADP, adenosine diphosphate; AMP, adenosine 5′-monophosphate: ATP, adenosine triphosphate.

nucleotidase activity was low already early in the rest phase, 3 h after lights on, and remained low until the end part of the rest phase, 8 h after lights on. Endo-5′-nucleotidase lacked diurnal variation.[23,24] These activity patterns may be associated with the lower level of energy metabolism during sleep as compared to wakefulness. Transport of adenosine through the cell membrane is the main pathway to remove adenosine from the extracellular space.[25] The transport is very effective: the half-life of adenosine is only 0.6–1.5 s. The concentrations of intra and extracellular adenosine are in equilibrium, which is maintained by nucleoside transporters. Two types of nucleoside transporters are known: equilibrative (ENT-family) transporters, which transport adenosine through the cell membrane in both directions according to the concentration gradient, and concentrating transporters (CNT-family), which are sodium dependent, and transport adenosine into the cell, also against the concentration gradient (for review, see[25]). Thus, in addition to substrate availability, the transport of adenosine through the cell membrane is an efficient site of regulation for extracellular adenosine concentrations. Under normal conditions the concentration gradient is such that adenosine is transported into the cell, but under circumstances when intracellular adenosine concentration increases, it can also be transported out of the cells.[26] If adenosine transport under normal conditions is inhibited by nucleoside transport inhibitors, e.g., S-(4-nitrobenzyl)-6-thioinosine (NBTI), extracellular adenosine concentrations increase.[27]

Several of the nucleoside transporters have been cloned, and the distribution of rat ENT1 and ENT2 mRNA in the brain has recently been described[28,29] as well as ENT 3[30] and ENT 4,[31] which are pH sensitive.

Extracellular adenosine can originate either from the neurones or the glia. The location of 5′-ectonucleotidase is associated more closely to the glia than the neurons,[32] pointing out the necessity of not overlooking the glia as a source of extracellular adenosine. Astrocytes are in excellent position to modulate large neuronal groups, since a single astrocyte can contact more than 100,000 synapses, and they release ATP that is metabolized extracellularly to adenosine.[33] ATP is also co-released with several neurotransmitters and metabolized to adenosine by 5′-ectonucleotidase. If the intracellular concentration of adenosine increases, e.g., during hypoxia, the concentration gradient will push adenosine from the intracellular space to the extracellular space through the ENT transporters.[26] We have reported previously that some astrocytes showed ENT1 staining, while ENT2 staining on astrocytes was observed only sporadically.[29] An additional possible source of extracellular adenosine is cyclic AMP (cAMP), released through noradrenergic stimulation.

Adenosine receptors

The effects of adenosine on neuronal activity are mediated by specific receptors. Four types of adenosine receptors are presently known: A1, A2a, A2b and A3. A1 and A2 receptors have opposite effects on the activation of adenylyl cyclase and formation of cAMP: A1 and A3 receptor activation decreases while A2 receptor activation increases cAMP. Through A1 receptors the effects of adenosine are inhibitory in the target tissue, through A_2 receptors excitatory. Adenosine A1 receptors are located both pre and postsynaptically. The presynaptic action of adenosine is mediated through G-proteins that are coupled to inhibition of N-type Ca^{2+} channels and stimulation of K^+ channels. While A1 receptors are widely distributed in the CNS, A2 receptors are located mainly in the striatum, nucleus accumbens and olfactory bulb. Lower amounts of A2a receptors have been found also from other brain areas, e.g., VLPO (for review, see[34]). Small amounts of A_3 receptor are found in the CNS (for review, see[35]).

The A1 receptor is coupled to the Gi_3 subtype of G-protein, which is known to either inhibit adenylate cyclase or activate phospholipase C (PLC) as effector pathways.[36] Since during SD the extracellular levels of adenosine increase two-fold in both cats and rats in the BF area,[27,37] it is possible that the effects of higher levels of adenosine could also be mediated by activating PLC and subsequently protein kinase C (PKC).[37]

Energy metabolism and sleep

Introduction

According to the classical view, brain energy supply is based on the continuous availability of glucose, which is the main source of brain energy metabolism. To a lesser extent energy is available from glycogen, which is stored by astrocytes. There is a close relation between the activity level of neurons and energy consumption in the brain: as neuronal activity increases, so does energy consumption.[38] Adenosine is the end-product of hydrolysis of the universal “energy currency”, ATP, thus energy consumption and adenosine formation are directly related.

It is safe to postulate that energy metabolism is somehow involved in the regulation of sleep, since all normal brain functions require undisturbed energy metabolism. Several indirect lines of evidence indicate that energy metabolism decreases during sleep.[39] Whether energy metabolism has a direct role in sleep regulation remains questionable, although many experiments support this notion. The theory put forward by Benington and Heller[40] stated that depletion of brain glycogen stores during (prolonged) wakefulness leads to increase in sleep need. Supported by a number of experimental data and drawing the attention to astrocytes in regulation of sleep (glycogen is almost entirely stored in astrocytes) this theory stimulated a lot of research with interesting, but controversial, results.[41,42] However, it is now apparent that the relations between sleep and energy metabolism are far more complicated than originally suggested. The glycogen theory and experiments that led to its development have recently been reviewed.[43] The following section introduces some new experiments to explore this question.

Lactate and sleep

A more recent hypothesis suggests that neurons can use lactate as an energy source and even favor lactate over glucose. During times of activation the utilization of lactate as a source of energy for neuronal activation has been supported by a number of experiments (for review, see[44]), although its limitations have also been introduced.[45] The core of this hypothesis is that upon neuronal activation, astrocytes metabolize glucose to lactate, which is then transported to neurons via selective transporters and that neurons use it as energy substrate instead of glucose. As the lactate that neurons utilize is produced by astrocytes and is transferred to neurons via the extracellular space, *in vivo* microdialysis measurement of lactate can be used as a marker of neuronal activation. We have shown in several rodent experiments that during SD, lactate concentration in the BF (SI/HDB/MCPO) increases.[46,47]

In humans we showed that extracellular lactate increased during brain activation in young subjects, but after SD this increase was either reduced or absent, as was the case in old subjects even without SD.[48] Similarly, in old rats lactate levels did not increase as compared to young rats in response to SD, indicating that the regulation of fast energy availability is disturbed during aging.[49] Importantly, we were able to show that increase in neuronal activation through NMDA receptors is required to induce the increases

in adenosine and lactate in BF during SD and increase in subsequent recovery sleep.[10]

Experiments from other laboratories have also produced data to support the idea that SD affects lactate production and transport in the brain: glucose transporter 1, an astrocyte-associated glucose transporter, was upregulated during SD.[14] Moreover, lactate dehydorgenase 2B protein, also associated with astrocytes, was increased with SD,[50] indicating that astrocyte–neuron energy metabolism circle that utilizes lactate is activated during prolonged waking.

AMP-activated protein kinase (AMPK) and sleep

The AMP-activated protein kinase (AMPK) acts as cellular metabolic sensor: when activated by increases in cellular AMP it promotes energy restoration and inhibits energy consumption. In brain, AMPK is activated by metabolic stress or neuronal activity-mediated rise in intracellular Ca^{2+}.[51] We have recently shown that AMPK is activated (phosphorylated) in the BF in the course of SD in rats.[52] Preliminary results from another laboratory showed also higher AMPK phosphorylation after SD in the BF and cortex[53] but a later publication from the same group announced technical difficulties in the measurement.[54] The level of phosphorylated AMPK was reported to increase after SD in the hypothalamus,[55] while pharmacological inhibition/activation of AMPK by ICV infused agents decreased and enhanced delta power respectively.

Neuronal activation and induction of (recovery) sleep

If we postulate that increase in neuronal activation is causal for induction of (recovery) sleep, we need to be able to show that lack of this activation results in decrease in sleep. We used theta/delta ratio as an EEG marker of neuronal activity, and lactate as a metabolic marker of neuronal activation. Using these measures we showed that increase in neuronal activation during waking is needed to induce sleep.[10] We induced excessive waking using two glutamate agonists: NMDA and AMPA. Of these NMDA prolonged waking and induced similar EEG spectral profile (characterized by increase in θ frequency) as SD, while AMPA receptor stimulation prolonged waking as much as NMDA, but did not increase θ activity. Adenosine and lactate were increased by NMDA, while AMPA did not increase them, and increase in delta sleep was only induced by NMDA. Interestingly, prolonging wakefulness in aging rats induced less theta, less adenosine and less lactate, as well as less recovery sleep[49] indicating that the sleep disturbances observed in aging rodents may originate from their inability to maintain high level of neuronal activity during waking.

Energy depletion and sleep

A direct relationship between energy depletion and sleep was assessed in an experiment where energy depletion was locally induced by infusing molecules that interrupted the electron transfer chain in the mitochondria. The hypothesis was that energy depletion would increase extracellular adenosine concentration and sleep. Indeed, 3 h infusions of an inhibitor of mitochondrial electron transfer chain 2,4-dinitrophenol (DNP) increased both extracellular adenosine concentration and sleep, when infused into the BF. Interestingly, when infused outside BF, extracellular adenosine concentration increased equally, but sleep remained unaffected.[47]

Summary and conclusion

In summary, in a series of experiments we explored the relationship between neuronal activation, energy metabolism and sleep. We showed that NMDA receptor mediated increase in neuronal activity increases extracellular lactate and adenosine concentrations in the BF, and subsequently sleep is increased. We also showed that experimentally induced energy depletion in the BF (but not outside) increased sleep. Moreover, the energy sensor, AMPK was phosphorylated in the course of SD. These results give support to the idea that an increase in sleep need is associated with increased neuronal activity and deterioration of energy balance. Based on these data we have formulated a hypothesis that continuous neuronal activation of the BF neurons during prolonged wakefulness, mediated mostly via NMDA receptors, will increase the energy demand and will lead to lactate release from astrocytes, as well as increases in extracellular adenosine. This sequence of events is central in the induction of recovery sleep.

Adenosine levels during sleep–wake cycle and effects of adenosine levels on sleep

Adenosine levels during spontaneous sleep–wake cycles

The measurements of extracellular adenosine concentrations during spontaneous sleep–wake cycles during lights-on period were made in several brain areas of cats using *in vivo* microdialysis.[27,56] Basal levels of adenosine measured in microdialysate samples from probes in the BF and thalamus ($N = 10$) were 32.8 ± 3.0 nmol/l (mean ± SEM); hence, from these data we estimated that extracellular adenosine concentrations are in the range of 165–300 nM, based on a *in vitro* microdialysis probe recovery of 10–20%. This approximation is in good accordance with a previous estimate presented by Ballarin et al.[57]

Adenosine levels in samples collected during spontaneous waking were higher than during periods of NREM sleep for all structures studied to date. In cats, the different brain regions had approximately the same reduced level in NREM sleep, which was approximately 75–80% of waking values. Since wakefulness in freely moving animals includes motor activity the question might arise as to which part of the differences in adenosine concentrations might be due to neuronal activity related to movement. An earlier work reports higher extracellular adenosine levels in the rat neocortex during the dark period (active period of rats) than during the light period,[32] a finding that was interpreted as evidence of association of adenosine levels with motor activity. Interestingly, in our studies the brain region most directly linked to movement, the ventral anterior/ventral lateral (VA/VL) nuclei of the thalamus, did not show a sustained rise in adenosine levels with SD, unlike the BF, which is an area less directly linked to movement. Thus it is unlikely that the regional differences in adenosine levels observed during SD and recovery sleep simply reflect movement-related changes.

In rats, adenosine levels in the BF were measured during lights-on period and found to be 12–15 nM.[37] Another group has reported diurnal variation in adenosine levels, being at the level of 60 nM during lights-on period and about 70 nM during lights-off period.[58] Nocturnal elevation of adenosine in the BF was also described by McKenna et al.[59]

Adenosine levels during prolonged wakefulness

Measured using *in vivo* microdialysis in cats as described above, adenosine concentrations increased in the BF steadily in the course of 6 h SD, reaching at 6 h twice the values that were measured in the beginning of the SD period[27,56] (Fig. 2). During the recovery sleep period, after the deprivation ended, adenosine levels in the BF gradually decreased over the next 2–3 h, a time when cats were mainly in NREM sleep. However, when samples were collected in different brain regions during 6 h total SD we found quite different patterns. The rise in adenosine levels in the cortex site paralleled

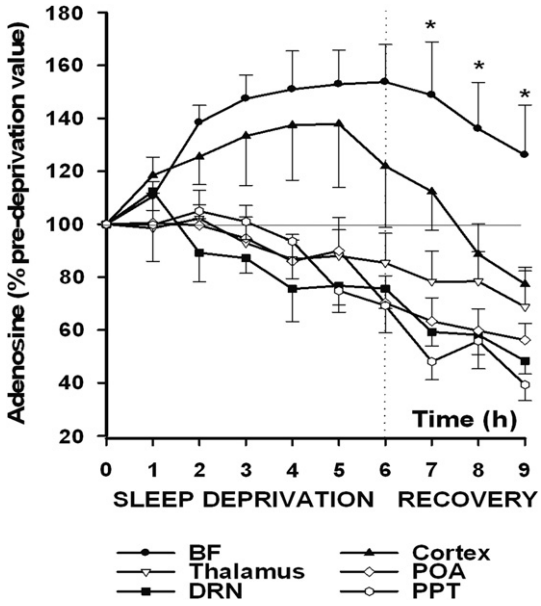


Fig. 2. Adenosine concentrations in different brain areas of cats during SD and recovery sleep. Cats were kept awake for 6 h. Baseline wakefulness values were measured from samples collected before the SD (pre-deprivation value = hour 0; defined as 100%). Two patterns were evident in the SD-induced changes in adenosine; in the BF and cortex adenosine levels increased during the SD, whereas in the other 4 areas adenosine levels were either stable or declined slowly during the 6 h of SD. During recovery sleep adenosine concentrations were significantly higher in the BF than in all other areas. Significant differences in adenosine concentration between BF and all other areas are marked with asterisks. Abbreviations: DRN, dorsal raphe nucleus; POA, preoptic hypothalamic area; PPT, pedunculopontine tegmental area (modified from Ref.[56]).

those in the BF during the first 5 h, although the rise was not as steep as in the BF, and started to decline in the last hour of deprivation. In the VA/VL of thalamus, dorsal raphe nucleus, PPT and POAH adenosine levels did not rise during 6 h SD, though many of these regions are also known to be active in sleep regulation. Interesting regional differences in extracellular adenosine concentrations were also found during the recovery sleep that followed SD. During the 3 h recovery sleep period, adenosine concentrations in the BF declined slowly, remaining significantly higher than the pre-deprivation basal wakefulness level at 3 h of recovery sleep. In all other brain areas the adenosine concentrations were either at the level of pre-deprivation values, or lower throughout the recovery sleep period.

We have repeated the key experiments in rats, and shown that extracellular adenosine concentrations in the BF of this species increases during SD from 12 nM to 25 nM within 3 h of deprivation.[37] Several later studies from our group[46,47,49] and from other groups[58] have confirmed, using gentle handling technique, that adenosine increases in the BF in the course of SD in young rats. Gradual increase in adenosine has been shown in rats also in the course of 30 h of sleep interruption protocol modeling sleep fragmentation during sleep apnea in humans.[59]

Interestingly, aging decreases the adenosine response to SD, although the baseline adenosine levels are higher in old animals.[49,58] An increase in adenosine concentration from 60 nM up to about 140 nM in young and from 120 nM up to 190 nM (approximately 50% increase) was reported by Murillo-Rodriquez et al. Results from our laboratory show even less increase, only about 20%[49] indicating that the normal adenosine response to sleep restriction is attenuated in the old animals, and perhaps explaining at least partially the attenuation in the homeostatic sleep response in old animals. Interestingly, adenosine receptor binding decreased with age in humans, indicating A1 receptor loss in the course of aging.[60]

Early pharmacological studies

Early studies by Radulovacki et al. showed that ICV infusion of adenosine and adenosine analogs increased sleep and reduced wakefulness.[61] Also systemic, as well as ICV administrations of A1 receptor agonists induced sleep.[62,63] Furthermore, the well-known stimulants of wakefulness, caffeine and theophylline, have since been determined to be antagonists of adenosine receptors.[35] More recent work has focused on identifying the specific brain structures and receptor subtypes on which adenosine acts to induce drowsiness and sleep.[64]

Change in adenosine metabolism/transport affects sleep

Also other substances that modulate extracellular adenosine levels can modify sleep. Inhibition of both enzymes metabolizing adenosine (Fig. 1) – AK[65] and ADA[66] – increase sleep. A functional polymorphism of the ADA gene in humans is associated with inter-individual variability in sleep architecture and the sleep EEG.[67] Both the duration and intensity of slow-wave sleep were increased in individuals with ADA G/A genotype as compared to G/G genotype.[67] The polymorphism (G to A transition at nucleotide 22) is functional since the enzymatic activity of the G/A variant ADA is 20–30% lower than that of the G/G variant, presumably leading to higher tissue adenosine levels. When ENT1 type adenosine transporter was blocked by NBTI, extracellular adenosine concentration increased.[31] When NBTI was locally infused to BF of both cats and rats, waking was decreased and both NREM and REM sleep were increased.[27,68]

Summary and conclusions

In summary, the data indicate that during spontaneous sleep extracellular adenosine concentration uniformly decreases in several brain regions, while during prolonged wakefulness adenosine increases selectively in the BF and, also, later and to a lesser extent, in the cortex. This unique pattern of regionally-specific rises in adenosine in response to SD supports the hypothesis that adenosine promotes the transition from wakefulness to slow-wave sleep by inhibiting the discharge activity of wake-active cortically projecting neurons of the BF, and possibly also by inhibiting neurotransmitter release from the cortical terminals of these BF neurons.

Which receptors mediate the actions of adenosine on sleep?

Introduction

Several lines of evidence indicate that the effects of adenosine on recovery sleep are mediated mostly through A_1 receptors (A1R, for review, see[64]), though also evidence about a role for A2a receptors (A2aR) has emerged. Interestingly, the effect of caffeine, the non-selective adenosine receptor antagonist, is thought to be mediated through A2aR.[69,70] Below we discuss the effects of genetic modifications of A1R and A2aR as well pharmacological activation/blocking of these receptors.

Sleep in adenosine receptor knock-out (KO) mice

Sleep recordings in three types of adenosine receptor KO mice have so far been reported: constitutive A1R KO mouse,[71] A2aR KO mouse[69] and conditional A1R KO mouse.[72] The constitutive A1R KO mice had normal sleep duration and sleep homeostasis, while they were unresponsive to A1R agonist.[71] One possible explanation for the lack of effect on sleep is compensation of function in a constitutive preparation. This view is supported by the finding that in conditional A1R KO mice sleep homeostasis is impaired.[72]

Interestingly, the conditional A1R KO mice were able to express slow-wave activity in baseline condition, but they were unable to increase it to compensate for sleep loss after SD. This finding emphasizes the role of adenosine and A1R in sleep homeostasis, perhaps rather than spontaneous sleep–wake cycle.

A2aR KO mice have lower heart rates, body temperature (males) and locomotion activity as compared to A1R KO and wild-type littermates,[73] as well as abnormal autonomic control during REM sleep. When A1R agonist was injected into the lateral ventricle of the KO animals, sleep–wake/cycle was not modulated, leading authors to suggest that this function was due to A2aR.[74] Alternatively, the absence of effect could be explained by using very low dose of A1R agonist since it did not affect sleep of wild-type mice either. However, under baseline conditions circadian sleep–wake profiles were identical for A2aR KO, A1R KO and their respective wild-type littermates.[69]

Evidence to support a role for A2aR in caffeine-mediated increase in waking comes from a study with A2aR KO mice. In this study caffeine was administered to mice that had been genetically engineered so that they either had no A2a or A1R, as well as to normal mice with both receptors intact. Caffeine increased waking in the wild-type mice as well as in A1R KO mice, but not in A2aR KO mice, indicating that the effects of caffeine are mediated through A2aR.[69] However, it is still possible that this effect is secondary to the effect of caffeine on locomotion: caffeine at low and medium doses has been shown to increase locomotion within 3 h after injection in wild-type and A1R KO mice.[73] In A2aR KO mice this effect is attenuated or even absent.[73] The role for A2aR in mediating the effects of caffeine is supported also by human data: A single c.1083T > C polymorphism in the A2aR gene (ADORA2a) modulates individual sensitivity to subjective and objective effects of caffeine on sleep.[70] However, a recent publication shows that caffeine and A1R antagonist infusions increased acetylcholine release in prefrontal cortex of mice and induced arousal, while A2aR antagonist did not.[75] As increase in cortical acetylcholine level is generally associated with increased waking, this experiment implicates that the effects of caffeine in prefrontal cortex may be mediated through A1R.

Interestingly, a study conducted on fruit flies shows that chronic administration of caffeine reduces and fragments their sleep,[76] while mutants lacking the fly adenosine receptor (with sequence similar to mammalian A2aR) had normal amounts of baseline sleep, as well as normal homeostatic responses to SD. Surprisingly, these mutants respond normally to caffeine. However, a phosphodiesterase inhibitor, IBMX (3-isobutyl-1-methylxanthine) mimicked the effects of caffeine in the mutant flies, while the effects of caffeine on sleep were blocked in flies that had reduced neuronal phosphodiesterase (PDE) activity. The authors conclude that chronic administration of caffeine promotes wakefulness in *Drosophila*, at least in part, by inhibiting cAMP PDE activity, but not through adenosine receptors. Whether such mechanisms contribute to wakefulness induced by chronic caffeine intake also in mammals remains to be clarified with further studies.

A1 receptors

A series of experiments examined the role of A1R in mediating adenosine effects in the BF using *in vivo* microdialysis. Local blocking the A1R using Cyclopentyltheophylline (CPT) decreased sleep and increased wakefulness,[77] the effect opposite to that produced by adenosine infusion.[78] Inactivation of the A1R using an antisense oligonucleotide targeted at the A1R gave the same outcome: sleep was decreased and waking increased.[79]

The effects of SD on subsequent recovery sleep can be decreased by blocking the A1R during the deprivation, either pharmacologically[80,81] or using A1 specific antisense.[79] Also, we showed that, in contrast to A1R antagonist effects, blocking of A2aR did not affect recovery sleep.[80]

A2 receptors

Several studies also suggest a role for A2aR in regulation of vigilance states, and that caffeine may act through these receptors when reducing sleep. The key structure in this action appears to be the subarachnoid space, below the rostral BF. In this area prostaglandin D2 receptor activation is followed by an increase in the concentration of extracellular adenosine, leading to induction of sleep through activation of A2aR (for review, see[82,83]).

Pharmacological studies showed that infusion of the A2a agonist into the subarachnoid space increased slow-wave sleep. This effect was blocked by the A2a antagonist.[82,83] Injection of the selective A2aR agonist into the subarachnoid space mimicked the sleep-promoting effects of prostaglandin D2, whereas an A1R agonist did not.[82] The authors state that the effects were likely mediated via A2aR in the tuberculum olfactorium/ventral nucleus accumbens.

Sites of action and cellular mechanisms for adenosinergic regulation of sleep

Introduction

As mentioned above, depending on the brain area, the adenosinergic regulation of sleep is predominantly mediated either via A1R or A2aR. A number of studies have addressed the electrophysiological mechanisms by which adenosine modulates the neuronal activity that regulates vigilance states. The induction of sleep by adenosine can take place either by inhibiting waking-active cells, or by stimulating the sleep-active cells, or both.

The basal forebrain

In vitro studies have shown that in the BF adenosine decreases neuronal activity through A1R by acting on inwardly rectifying K^+ channels or the hyperpolarization-activated cation current. Both cholinergic and non-cholinergic neurons are inhibited by adenosine, but this inhibition is mediated by different ion currents: cholinergic neurons through inwardly rectifying K^+ current and the non-cholinergic neurons through a hyperpolarization-activated cation current (for review, see[84]).

Does adenosine inhibit both wake-active and sleep-active BF neurons equally? Increasing adenosine concentration in the BF, or infusing adenosine A1R agonists decreased the activity of wake-active neurons, while the A1R antagonists increased it.[85,86] An A2aR receptor agonist had no effect on wake-active neurons,[86] indicating that the decrease in neuronal activity in this area was mediated through A1R. Neuronal activity of the sleep-active cells was reported to decrease[85] or not to change[86] in response to application of A1R agonist or adenosine.

The BF wake-active cholinergic neurons are tonically inhibited by adenosine through A1R, and the removal of this inhibition by blocking these receptors would increase their firing as well as cortical arousal. The activity of the cholinergic neurons can be regulated by adenosinergic manipulations: both theophylline and caffeine can modify acetylcholine levels and metabolism in the brain.[87] Caffeine increases cortical acetylcholine dose-dependently and already at doses that can be induced by ordinary caffeine consumption.[88]

Evidence that directly supports the hypothesis that the effect of adenosine on recovery sleep in the BF is based on the inhibition of

the cortically projecting cholinergic cells was provided by a series of experiments where we showed that specific destruction of the cholinergic cells in the BF prevented increase in extracellular adenosine during SD and abolished recovery sleep.[89] To destroy the cholinergic cells, we used 192 IgG-saporin (saporin), toxin that is specifically binds to p75 nerve growth factor receptors, which lie only on BF cholinergic cells.

Unfortunately, a lot of confusion has been created by results published by another research group. In this report the authors stated that ICV saporin lesion did decrease adenosine levels during SD but that recovery sleep remained normal.[90] This conclusion was most probably based on an inaccuracy in experimental planning: the effect of saporin lesion on adenosine levels and on recovery sleep were measured in separate groups of animals, and very regrettably, at different time points: recovery sleep was measured at 2 weeks after the lesion, while adenosine levels and the number of cholinergic cells in the BF were measured at 3 weeks after the lesion.[90] We showed that after ICV administration, the saporin lesion is complete only after 3 weeks,[89] indicating that at the reported time point of measurement of recovery sleep response, the cholinergic cells were not destroyed, explaining the intact recovery response. Also another group, consistent with our data, has reported that locally induced saporin lesion in the BF decreased recovery sleep.[91]

The hypothalamus

Hypothalamus is another important site in adenosinergic modulation of sleep. Adenosine, acting on A1R, promotes NREM sleep by inhibiting the histaminergic neurons in the tuberomammillary nucleus.[92] Interestingly, in addition to A1R-mediated modulation, adenosine A2aR agonist has also been reported by the same group to increase sleep by inhibiting histaminergic system.[93] This effect appeared to be mediated through induction of GABA-release in the tuberomammillary nucleus.

In perifornical-lateral hypothalamic area (PF-LHA), where the wake-promoting hypocretin neurons are located, A1R agonist significantly suppressed arousal and produced NREM and REM sleep, while A1R antagonist produced arousal and suppressed NREM and REM sleep.[94] A further study showed that an A1R agonist suppressed sleep–wake discharge activity of PF-LHA neurons, and that A1R agonist and antagonist suppressed and increased, respectively, c-fos immunoreactivity in both hypocretin and non-hypocretin neurons.[95] This was in agreement with *in vitro* data showing that adenosine via A1R inhibits hypocretin neurons[96] as well as the glutamatergic synaptic transmission and plasticity in hypocretin neurons.[97] Altogether, these results demonstrated that the wake-promoting system is subject to increased endogenous adenosinergic inhibition and that adenosine acting via A1R, in part, inhibits hypocretin neurons to promote sleep.[95]

Infusion of A1R agonists, or increasing extracellular adenosine concentration by NBTI in the lateral preoptic area (LPO), where the sleep-active cells lie, induced waking, while A2aR stimulation induced sleep.[68] Similarly, in the VLPO, adenosine, through A2aR, has been shown to excite the GABAergic, sleep-inducing neurons and promote sleep.[98] It was suggested that the ability of adenosine to activate sleep-active POAH/VLPO neurons is through an indirect mechanism: by blocking GABA inhibitory inputs, adenosine may disinhibit these neurons and promote sleep.

Cortex and the thalamocortical system

In the thalamocortical system, adenosine, acting through A1R, facilitates cellular oscillations at slow frequency after SD (for review, see[84]). In thalamocortical neurons, adenosine postsynaptically decreases the I_h current and induces hyperpolarization by increasing the G-protein-dependent inwardly rectifying K^+ channel conductance; both of these events depress activating influences from the brainstem and promote burst discharges.[99] Presynaptically, adenosine's inhibitory effects can lead to functional deafferentation of the thalamocortical circuits resulting in increased influence of inwardly rectifying K^+ channels and hyperpolarization-activated current to facilitate delta activity.

During the latter part of 6 h SD in cats, adenosine levels increased also in the cortex, but started to decline already before the end of the deprivation.[56] Interestingly, a recent report on SD in rats shows that cortical firing rates of neurons increase up to 3 h of SD, and then reached a plateau.[100] Neuronal firing rates increase during SD also in the BF of rats.[101] These increases in neuronal activity may, at least partially, explain the increases in adenosine levels.

Administration of adenosine to prefrontal cortex also modifies vigilance state. Generally, administration of A2a agonists increased wakefulness (including increase in acetylcholine release) and A1R agonists reduced it. In contrast, specific A1 antagonist or caffeine increased acetylcholine release and waking, while A2a antagonist was ineffective.[75] Interestingly, administration of adenosine agonists and antagonists in the prefrontal cortex modulated acetylcholine release in the pontine reticular formation, suggesting that adenosine, via A1R, is able to modulate behavioral arousal via descending input to the pontine brainstem.[75]

The pons

Also the pons region, which is the central regulator of REM sleep, is affected by adenosine. Microinjection of adenosine A1 receptor agonist or an inhibitor of adenylyl cyclase into the caudal, oral pontine reticular formation (PnOc) of the rat induces a long-lasting increase in REM sleep.[102] On the other hand, an adenosine A2a receptor agonist increased acetylcholine release in the pontine reticular formation (PRF) and both NREM and REM sleep.[103] The authors conclude that the enhancement of REM sleep is consistent with the interpretation that adenosine A2a receptors in the PRF contribute to REM sleep regulation, in part, by increasing acetyl choline (ACh) release in this area. A2a receptor activation may promote NREM sleep via GABAergic inhibition of arousal promoting neurons in the PRF.[103]

Role for glia?

Extracellular adenosine can originate either from neurons or from glia. A role for astrocytes as source for extracellular adenosine is suggested by experiments that used genetically engineered mice, whose glia transmission was impaired by the expression of a dominant negative SNARE domain in astrocytes. It was found that these mice cannot release ATP from astrocytes leading to decreased extracellular adenosine concentration.[33] These animals spent the same amount of time in waking, NREM, and REM sleep as wild-type mice. However, the mutant mice showed reduced slow-wave activity and a decrease in recovery sleep following SD.[81] As A1R antagonist suppressed homeostatic sleep response in wild-type mice but was not able to suppress sleep in the mutant mice, the authors conclude that astrocytes modulate the accumulation of sleep pressure via a pathway that involves A1R. Whether the vesicular release of ATP from astrocytes is connected to regulation of neuronal energy metabolism is not presently clear.

Summary and conclusion

It is thus clear that adenosine modulates vigilance states through many different pathways. The key areas so far have been found in the BF/hypothalamus region and pons, where also the classical centers for vigilance regulation are situated.[2] It thus appears that adenosine mainly works through the global regulatory networks to induce (recovery) sleep. However, the increase in adenosine concentration during SD also in the cortex may indicate that adenosine has a role in the local regulation of sleep and in addition, may regulate vigilance also through descending routes.

Prolonged wakefulness and adenosine

Effects of prolonged wakefulness on adenosine metabolism and transport

We found no changes in the activity of AK, ecto-5′-nucleotidase or endo-5′-nucleotidase after 3 or 6 h of SD either in the BF area or in the cortex.[23] Similar results have been reported by another group that measured the activities of both the cytosolic and extracellular forms of 5′-nucleotidase,[104] as well as activity of ADA and AK. Enzymes were found to exhibit diurnal variation in many brain regions, while SD did not alter the activity of any of these enzymes.[104] These studies suggest that modulation of the activities of adenosine enzymes is not a likely mechanism to explain the increase in adenosine that occurs during wakefulness.

Another obvious site of adenosine concentration regulation during SD is the transport across cell membrane. Studies of nucleoside transport are difficult, since no specific antibodies, or specific agonists or antagonists are available. We have used tritium labeled NBTI-binding as marker of ENT1 transport activity. The activity of the transport was decreased after 6 h of SD in the BF but not in the cortex.[105] This decrease may contribute to the increase of extracellular adenosine, but is highly unlikely to fully explain it.

Could high extracellular adenosine concentrations induce long-term effects in neurons?

While many of the somnogenic effects of adenosine may represent direct receptor mediated actions, there are long-duration effects of SD whose time course suggests that they are likely to be mediated through effects on transcription. For example, it is known that in humans successive nights of sleep restriction have a cumulative effect on waking neurobehavioral alertness[106] and that the effects of prolonged SD in animals[107] or humans[106] are not fully recovered during a single night of recovery sleep. The extracellular levels of adenosine start to decrease as soon as sleep is initiated. Though the decline in the cat BF after 6 h of prolonged wakefulness was not completed during 3 h of recovery sleep, the time course of the decrease to control adenosine levels is probably of the order of hours, not days. The extension of the effects of prolonged wakefulness over several days suggests that intracellular events, including protein synthesis, may be activated. Adenosine induces, through A1 receptor mediated increase in intracellular calcium, the activation of the nuclear translocation of NF-kappaB, which occurs almost exclusively in the cholinergic neurons of the basal forebrain following 3 h of SD.[108]

SD increases A1R binding in the cortex of humans[109] as well as A1R density in rats.[108] The increase in extracellular adenosine concentration thus is paralleled by increasing receptor number, creating an effective increase in the signal. Although the time span of this increase has not been addressed, it clearly expands the duration of the adenosine signal beyond the limits of the metabolism of the adenosine molecule. This enhancement might produce a hyper responsiveness to adenosine, which in turn, might lead to the vigilance and performance decrements observed in "sleep debt", even at adenosine levels that do not normally impact alertness and performance.

Adenosine and cognitive function

Conditional A1R KO animals showed reduced slow-wave activity as response to SD and during the restriction period they failed to maintain normal cognitive function, although this function was normal when sleep was not restricted.[72] Alterations in memory performance have been reported in another genetically engineered mouse strain with impaired gliotransmission[81]: in these animals novel object memory task was impaired in wild-type animals after SD, but not in the transgenic animals. Infusion of A1R antagonist in wild-type animals induced similar behavior as in transgenic animals.[81] When adenosine concentration in the BF of rats was increased using *in vivo* microdialysis, their vigilance performance was decreased.[110]

At cellular level adenosine reduced long-term potentiation (LTP) in chronically sleep restricted rats (sleep restriction was induced by lesion of VLPO). LTP was partially restored when adenosine antagonists were introduced to the slices, implicating that accumulation of adenosine in the VLPO-lesioned animals could contribute to the LTP impairment in these animals.[111]

These data implicate that the behavioral and cognitive deficits that accompany SD are at least partially mediated by adenosine.

Beyond adenosine

An adenosine-associated sleep factor: NO

Several molecules and groups of biologically active substances have the ability to influence vigilance states. Several cytokines (for review, see[112]), brain-derived neurotrophic factor (BDNF),[14,113] GnRH,[114,115] chaperone protein BiP/GRP78,[116] PDG2[82] are among substances that clearly have sleep-promoting functions.

Recently a new candidate to this group was introduced: NO.[117–119] NO is a gaseous neurotransmitter that is involved in regulation of circulation and acts also in many pathological conditions (for review, see[120]). Increase in tissue concentration of NO increases adenosine concentration.[121] In a series of *in vivo* microdialysis experiments Kalinchuk et al.[117,118] showed that NO extracellular concentration in the BF (measured as cumulative increase in NO metabolites, nitrate and nitrite) increases in the course of SD, very similarly than that of adenosine (Fig. 3). Pharmacological increase of NO in the BF resulted in increase in sleep that was of the same magnitude as that observed after SD. When elevation of NO was prevented during SD, recovery sleep was abolished. It was further shown, that increase in NO during SD was mediated by the enzyme inducible NO-synthase (iNOS), that normally is not present in brain but is synthesized as response to immunological challenge. Interestingly, NO and adenosine concentrations appeared locked together: any manipulation that decreased NO decreased also adenosine, suggesting that the effects of NO on sleep may be mediated through adenosine. This hypothesis was tested using *in vivo* electrophysiological measurement of neuronal activity in the BF as response to pharmacological increase in NO concentration with and without pretreatment with adenosine A1 receptor antagonist. The firing rate change as response to NO induction was diminished with adenosine block, but not completely abolished,[119] indicating that at least part of the NO effects may be mediated through adenosine.

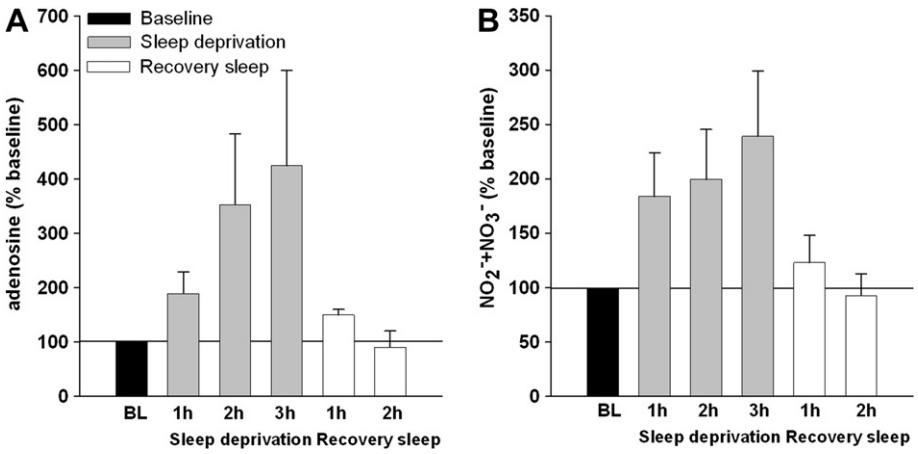


Fig. 3. Adenosine and NO concentrations in the BF of rats during SD. A) Adenosine concentration increases in the course of 3 h of SD (grey bars) and decreases within 2 h of recovery sleep (white bars). B) Cumulative measurement of nitrite and nitrate ($NO_2^- + NO_3^-$) as indicators of NO production shows increase in BF during 3 h of SD (grey bars) and decline during recovery sleep (white bars) (modified from[111]). Note the similar pattern of changes in adenosine and NO. BL (black bars) = adenosine or $NO_2^- + NO_3^-$ concentrations during pre-deprivation baseline.

The induction of iNOS by prolonged wakefulness may give light to the question whether induction of recovery sleep recruits additional regulatory mechanisms (substances) as compared to regulation of spontaneous sleep. As iNOS is normally absent in brain, its induction by SD may support the view that indeed, additional mechanisms exist. However, this question should be explored by more accurate iNOS measurements during spontaneous sleep–wake cycle.

Could adenosine be final common pathway for the action of various sleep factors?

Adenosine has biochemical properties that make it an ideal candidate to mediate the action of various somnogens. Adenosine is a ubiquitous and small molecule located both intra and extracellularly; the inhibitory A1R is also ubiquitous, as are the enzymes needed to produce and metabolize adenosine. The time course of adenosine cellular action is swift, as action at the A1R relies on neither gene transcription nor second messengers. In contrast, the clearance of adenosine from the extracellular space following the SD-induced increase of adenosine in the BF takes hours, suggesting it may carry the signal of "sleep need". In addition, high extracellular adenosine levels induce intracellular events (e.g., gene expression) that may further expand the time frame for the effects of adenosine.

Our hypothesis is that during spontaneous waking extracellular adenosine concentration increases modestly, via mechanism (s) that are independent of iNOS. During prolonged wakefulness, adenosine concentration, triggered by increased neuronal activation, increases in the extracellular space of the BF, increasing sleep propensity. iNOS-dependent increase in NO contributes to adenosine increase during SD. Adenosine inhibits the wake-active cells in the BF and promotes sleep. When the activity of the waking-active cells decreases sufficiently, sleep is initiated. During sleep the extracellular adenosine concentrations decrease and thus the inhibition of the waking-active cells also decreases allowing the initiation of a new waking period. The long-term effects of prolonged wakefulness are, at least partially, mediated by high extracellular adenosine concentrations that initiate intracellular signal transductions cascades resulting in gene transcription.

Two brain structures appear to be central in accumulation of sleep pressure: the cortex and the BF. Interestingly, adenosine increases exclusively in these areas during prolonged wakefulness.[56] Numerous experiments from other laboratories indicate that in the cortex there is a local, use-dependent accumulation of sleep propensity.[7,8] The role of adenosine in this process has so far remained under-explored. Our laboratory has contributed by showing that the level of cortical vigilance is a central factor in production of sleep propensity and recovery sleep.[10]

In addition to adenosine, other strong candidates for sleep factors have been introduced. However, evidence of ability to directly affect neuronal activity is strongest for adenosine, raising the question: could adenosine be the final pathway to modulate vigilance states through modulation of neuronal activity? Except for NO (see previous section), this question has not been systematically addressed. However, anectodical data give foundation for speculation that this may be the case. Prostaglandin D2, acting through DP1 receptors, releases adenosine in the vicinity of the basal forebrain,[122] forming the basis for prostaglandin-mediated modulation of vigilance. Interleukines (except IL-6) and TNF-alpha can induce iNOS,[123] and through iNOS then activate increases in adenosine concentrations. A growth hormone secretagogue (GHS, molecule that acts like GnRH in increasing GH secretion), improved sleep quality in man,[124] and adenosine has been shown to act as partial agonist for GHS receptor.[125] Facilitation of synaptic transmission and plasticity through neurotrophic factors, including BDNF, requires tonic adenosine A2a receptor activity.[126] Whether these connections between the suggested sleep factors and adenosine play a role in regulation of vigilance state, and particularly in sleep homeostasis, remains to be explored.

It must be emphasized that our model concentrates on the global effects of prolonged wakefulness, and on mechanisms that regulate this aspect of sleep regulation. We have identified a relatively specific area, the BF, which appears to be central in the regulation/execution of recovery sleep. Two sleep-inducing substances, adenosine and NO increase their concentration in this area during prolonged wakefulness, and increasing their concentration in this area induces sleep. Moreover, prevention of the increases abolishes recovery sleep.[80,117] The increase in adenosine and NO concentrations, as well as increase in sleep, is associated with intact cholinergic system since specific lesion of the BF cholinergic cells will abolish the increases in adenosine and NO, and also recovery sleep.[89] So far we have found no condition where the connection between BF adenosine (or NO) concentration and induction of (recovery) sleep would have been disrupted.

1. Practice points

- adenosine is related to both sleep regulation and energy metabolism
- adenosine is a physiological sleep factor
- adenosine modifies recovery sleep mainly in the BF area
- increase in adenosine concentration in the BF is dependent on intact cholinergic cells in this area

2. Research agenda

- to clarify at the cellular level the relations between energy metabolism and sleep
- to clarify the relationship and mechanisms of local vs. global regulation of sleep homeostasis
- to clarify the role of glia in regulation of recovery sleep

Acknowledgement

This work was supported by a grant from the Academy of Finland.

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